AI Usage Report

Name: Habibur Rahman Mahin
Roll Number: BSSE1422
Course: SE811 Software Maintenance
Assignment: Lehman's Laws of Software Evolution Analysis
AI Tool Used: Claude (claude.ai) by Anthropic

Introduction

This report describes how I used artificial intelligence as a learning and verification tool to complete the assignment on applying Lehman's Laws of Software Evolution to open-source project data (Linux Kernel evolution graphs).

Step 1: Self Study Through Theory

I began the assignment by going through the course lecture slides independently. I studied all eight of Lehman's Laws, their definitions, their conditions of applicability, and the examples discussed in class. This gave me a foundational understanding of what each law predicts about the behavior of evolving E-type software systems before I looked at any data.

Step 2: Independent Graph Analysis

After understanding the theory, I analyzed the provided graphs on my own without any AI assistance. I examined each figure carefully, identified visible trends such as monotonic growth, oscillation, bounded patterns, and cumulative complexity increases, and wrote down my own judgments about which laws I thought were applicable to each graph and why. This step was done entirely through my own reasoning and understanding.

Step 3: Using Claude AI for Verification

After completing my own analysis, I uploaded the class lecture slides and the assignment document to Claude AI on claude.ai. I asked Claude to analyze the same graphs and generate answers mapping each law to each graph with justifications. The purpose of this step was not to get answers to copy but to use the AI as a reference point to check the accuracy and depth of my own analysis.

Step 4: Comparison, Correction and Deeper Understanding

I then compared Claude's responses with my own written answers side by side. Where my answers differed from the AI's responses, I did not simply accept the AI's answer. Instead, I re-examined the graphs and re-read the relevant theory to understand whether the AI's reasoning was logically sound and whether I had missed something or misunderstood a concept. This helped me identify gaps in my understanding, such as subtle applications of Law III (Self-Regulation) and Law VIII (Feedback System) which I had initially found difficult to justify from graph data alone. Through this back-and-forth process I gained a much deeper understanding of how each law manifests in real project metrics.

Step 5: Final Report Preparation

After completing the study and re-analysis phase, I prepared the final assignment report based on my own corrected and refined understanding. The structure, reasoning, and final judgments in the report reflect my own learning process rather than a direct copy of AI-generated content. Claude was used as a study aid and verification tool, similar to how one might use a textbook answer key or discuss answers with a peer, not as a replacement for independent thinking and analysis.

Conclusion

The use of Claude AI in this assignment supported my learning process by allowing me to verify my work, identify errors, and deepen my understanding of Lehman's Laws through comparison and re-analysis. At no point did I submit AI-generated content without first understanding and validating it through my own reasoning. The AI served as an intelligent reference tool that enhanced the quality of my independent work.